When the Simulator Is Wrong, Do Better Derivatives Improve Decisions?

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Abstract

We use option pricing and hedging to study whether better surrogate derivatives improve decisions when the assumed simulator is wrong. A neural Heston surrogate predicts option values. A fixed hedging rule uses their sensitivity to asset price to choose an asset holding intended to offset changes in the option's value. At two Heston coefficient regimes excluded from training and validation, adding curvature labels to price-sensitivity supervision reduces root-mean-square error (RMSE) in the unsupervised variance sensitivity by 40 and 26%. Supervising all three targets reduces curvature and variance-sensitivity RMSE by 82–93% relative to two baselines without derivative labels. The incremental gains depend on the target and are assessed against measured solver disagreement. When simulated dynamics depart from the Heston model class, a value-only model captures at least 96% of each of the fully supervised model's eight improvements over the residual-only baseline at zero transaction cost. At the registered 1% cost, the fully supervised model is worse than that baseline in all ten departure settings. The preregistered confirmatory hedging endpoint, evaluated under a coefficient shift, also fails. These findings support cross-target transfer within the tested nested design, without establishing a consistent hedging advantage specific to derivative labels or a general ordering of label choices.

1 Introduction

Scientific surrogates approximate models that can themselves be wrong. We use option pricing and hedging to ask whether reducing derivative error improves decisions under model misspecification. The network predicts a Heston option value; a fixed hedging rule uses its asset-price sensitivity. Numerical references measure value and derivative errors; simulations outside Heston test whether these errors matter for hedging. A close value fit does not ensure accurate derivatives: at a held-out coefficient regime, two models without derivative labels have value errors of 1.6% and 3.3%, but curvature errors of 41% and 27% (Appendix D). A physics residual also leaves substantial derivative error, consistent with documented limitations (Krishnapriyan et al., 2021; Wang et al., 2021).

Derivative supervision is established in Sobolev training (Czarnecki et al., 2017) and differential machine learning (Huge and Savine, 2020). Molecular models fit forces (Ercolessi and Adams, 1994; Wang et al., 2019) or differentiate learned energies (Behler and Parrinello, 2007). Sobolev-PINNs augment residual training (Son et al., 2023). Our contribution is a controlled evaluation of which targets improve and whether this improves hedging. Model misspecification also affects simulation-based inference (Cranmer et al., 2020; Cannon et al., 2022). Sequential prediction also highlights the importance of evaluation distributions (Ross et al., 2011).

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Does fidelity translate under misspecification? We distinguish held-out coefficients within the assumed model from a generating process outside its class, which no parameter setting recovers. In the financial setting, this means asking whether more accurate Heston derivatives improve hedging when the simulated price process departs from Heston dynamics.

Where should the labels go? Derivative labels require additional reference computations. We add targets sequentially to measure their conditional contribution. Noguer i Alonso (2026) identifies the importance of second derivatives in discrete-time hedging but reports no hedging backtest. Malandain et al. (2025) reports inaccurate derivatives despite a small residual and proposes derivative regularisation. The closest trained configuration (Sakuma, 2026) supervises first and second derivatives together, but evaluates in distribution and reports similar pricing and Greek accuracy among its three supervised arms.

We compare a nested sequence of label sets, pre-registered controls, and numerical references with measured uncertainty. The registered tests and thresholds were fixed before training. Numerical cross-checks let us measure errors in the value and the evaluated derivatives, and assess whether differences exceed reference uncertainty.

What we find. Full supervision reduces held-out curvature and variance-derivative RMSE by 82–93% relative to the value-only and residual-only baselines. Adding curvature labels also improves the unlabelled variance derivative by 40 and 26%, but adjacent-rung gains depend on the target. Under model-class departure, value-only training largely reproduces the zero-cost hedging gains; at 1% cost the fully supervised model is worse than the residual-only baseline in all ten settings. The registered confirmatory test also fails (Sections 3.1 and 3.6).

What this is not. We do not correct the simulator's dynamics. We reduce the surrogate's error against an assumed pricing model and ask whether this improves a fixed hedging rule when the model is structurally wrong. The experiments test this distinction in one PDE setting; they do not establish transfer to other simulators or real-market performance.

2 Setup

Model and notation. The Heston model (Heston, 1993) describes an asset price S and its instantaneous variance v, both evolving randomly. The function $u(S, v, \tau; K)$ is the model value of a European call option: a contract giving its holder the right to buy the asset at a fixed price K (the strike) at expiry. With time to expiry τ, its payoff at expiry is $\max(S - K, 0)$. For fixed coefficients, the pricing PDE is

$$\begin{aligned} \partial_\tau u = \tfrac{1}{2} v S^2 \partial_{SS} u + \rho \xi v S \partial_{Sv} u + \tfrac{1}{2} \xi^2 v \partial_{vv} u \\ + (r - q) S \partial_S u + \kappa(\theta - v) \partial_v u - r u, \end{aligned} \tag{1}$$

with $u(S, v, 0; K) = \max(S - K, 0)$. Here κ is the variance mean-reversion rate, θ its long-run level, ξ the variance diffusion scale, and ρ the correlation between the price and variance shocks; r and q are the interest rate and dividend yield. The variance input is denoted v_0 in the data and configuration. The network approximates u across coefficients, as in parametric Heston PINNs (Hainaut and Casas, 2024); the squared PDE residual measures its failure to satisfy Eq. (1) and supplies the physics-informed loss (Raissi et al., 2019).

The three targets are *delta*, the price sensitivity $\partial_S u$; *gamma*, the curvature $\partial_{SS} u = \partial^2 u / \partial S^2$; and the variance sensitivity $\partial_v u$. The last is a derivative with respect to variance, rather than volatility. Delta describes how the option value changes with asset price; gamma describes how that sensitivity itself changes. All are input derivatives obtained by automatic differentiation, distinct from loss gradients with respect to network weights. The registered hedging rule uses delta to set its asset holding; gamma and variance sensitivity are supervision targets, not additional inputs to that rule.

Reference cross-checks. We cross-check option values using up to four solvers: Heston Fourier inversion (Heston, 1993; Albrecher et al., 2007), cosine expansion (Fang and Oosterlee, 2008), an alternating-direction finite-difference scheme (Craig and Sneyd, 1988), and Monte Carlo (Lord et al., 2010). The first two are separately implemented but share a characteristic-function formula, so they cannot exclude a common formula error. The predeclared mask combines agreement checks across values and derivatives, with a relative tolerance of 10^{-3}, a scale floor near zero, and an allowance for Monte-Carlo noise (Appendix B). It removes 17.1% of the interpolated grid and 9.2% of the

extrapolated grid, concentrated at short expiries and just below the strike. For each metric, all models are evaluated on the same retained points.

What one envelope is. Value agreement does not certify derivative accuracy. For each derivative and evaluation regime, an *uncertainty envelope* combines the largest pairwise solver disagreement with Monte-Carlo standard error (Table 4). A difference is called *resolved* when it exceeds that envelope; this is a resolution heuristic, not a statistical test or error bound. Unresolved differences need not be zero. Labels and reference share a solver family: holding out parameters prevents memorisation of those regimes but cannot exclude shared bias. Appendix B quantifies these limits.

Departure. An *anchor* is a fixed coefficient setting excluded, together with a neighbourhood, from both training and validation. Derivative accuracy is evaluated at two Heston anchors: one inside the sampled coefficient ranges and one with correlation outside them. These test interpolation into a held-out region and extrapolation within the model class. For departures outside that class, we evaluate hedging under Bates dynamics, which add price jumps (Bates, 1996), and Merton dynamics, which combine jumps with constant volatility (Merton, 1976). In these tests every model receives the same assumed Heston coefficients; no derivative-accuracy claim is made for the departed dynamics (Appendix C).

Arms. An *arm* is one training configuration. Both baselines use value labels: the value-only model has no PDE residual, whereas the residual-only baseline adds that residual but no derivative labels. A nested *ladder* adds labels to the residual baseline: rung 1 adds delta, rung 2 also adds gamma, and rung 3 also adds variance sensitivity. Each label loss is a scale-normalised squared error (Huge and Savine, 2020). These arms share architecture, optimiser, data and budget. The residual, gamma and variance-label weights were selected on validation; the delta-label weight was fixed. Adaptive loss balancing was not tested. The extra-value-data control uses a different residual weight, disclosed in Section 3.5.

The hedging rule holds $\widehat{\partial_S u}$ units of the asset to offset changes in the option's value. It updates that holding over 43 steps on $10{,}000$ shared simulated trajectories (Andersen, 2008). Hedging loss J is expected shortfall, the mean of the worst 5% of losses, so it measures adverse outcomes rather than average performance (Rockafellar and Uryasev, 2000); lower is better. Tests include zero and nonzero transaction costs. The pre-registered thresholds require a 10% confirmatory improvement, a 15% derivative-accuracy improvement, and specify a 2% negligibility floor (Table 3).

3 Results

We report the registered intervals as specified before training. Later comparisons are exploratory, without adjustment for multiple comparisons. Path-bootstrap intervals measure variation across simulated paths for the fitted models, not variation across new training runs.

Table 1: Held-out derivative RMSE on the full parameter grid, interpolated / extrapolated coefficient regime, five seeds; second-derivative entries $\times 10^{-3}$. Lower is better. Bold marks a lowest mean separated from the runner-up by more than the reference uncertainty; an unbolded regime–derivative cell is one whose top two arms this benchmark cannot separate. Resolution is the registered reference-noise heuristic, not a test. Lower block: those contrasts divided by the reference envelope for that derivative; italics mark a gap under one envelope, which this benchmark cannot resolve (Appendix B).

Arm	$\partial u/\partial S$	$\partial^2 u/\partial S^2$	$\partial u/\partial v$
residual-only baseline	$0.0537/0.0424$	$7.82/6.04$	$9.93/7.44$
value-only feedforward	$0.0881/0.0815$	$8.29/7.17$	$5.74/2.89$
rung 1, $+\partial u/\partial S$	$0.0106/0.0079$	$2.56/1.58$	$2.66/1.73$
rung 2, $+\partial^2 u/\partial S^2$	$0.0033/0.0034$	$1.02/0.64$	$1.59/1.28$
rung 3, $+\partial u/\partial v$	$\mathbf{0.0027}/0.0038$	$0.83/0.82$	$\mathbf{0.73/0.52}$
same contrasts, in reference envelopes			
baseline → rung 3	$184.7/97.3$	$9.1/2.6$	$85.0/54.5$
rung 1 → rung 2	$26.2/11.4$	$2.0/\mathit{0.5}$	$9.9/3.5$
rung 2 → rung 3	$2.2/\mathit{-0.97}$	$\mathit{0.2/-0.1}$	$8.0/6.0$

3.1 Derivative accuracy and hedging under model departure

Table 1 evaluates derivative accuracy at held-out coefficients within the Heston class. Relative to the two baselines without derivative labels, full supervision reduces gamma and variance-sensitivity RMSE by 82–93% and delta RMSE by 91–97%. These baseline-to-rung-3 improvements exceed the reference uncertainty at both regimes (Appendix B). The derivative-accuracy findings below concern these two regimes; departures outside the model class are evaluated through hedging loss.

At zero transaction cost, rung 3 has lower mean loss than the residual-only baseline in seven of nine Bates jump settings and in the Merton setting. The value-only model captures at least 96% of each of these eight gains and outperforms rung 3 in four. At the registered 1% cost, rung 3 has higher loss than the residual-only baseline in all ten settings. Thus the observed gains over that baseline do not establish a consistent hedging advantage specific to derivative labels.

Extent of misspecification. The Bates sweep varies jump arrival rate $\lambda_j \leq 0.5$ and jump-size standard deviation $\sigma_j \leq 0.15$ (Appendix C). These parameters order the tested departures, but we did not measure distance from the Heston model class. At zero cost the relative gain over the residual-only baseline decreases along each coordinate, from $+27.7\%$ at the in-class setting to -4.2% at the largest tested jumps, changing sign along the $\lambda_j = 0.5$ row. All six one-dimensional chains decrease in every one of five seeds, for both absolute and relative gains ($30/30$ each). This monotonicity analysis is exploratory and differs from the preregistered expectation of a non-monotone pattern. The registered departure-sweep verdict uses the 1% cost tier, where no favourable regime is found.

The registered derivative-accuracy endpoint also fails, despite gamma and variance-sensitivity reductions of 86.4–93.0% exceeding its 15% threshold. It additionally requires value accuracy to remain within a two-sided parity band; improving value error by 82–91% violates that band. We retain the registered failure and disclose this limitation of the criterion.

3.2 Adding curvature labels improves curvature accuracy

Adding gamma labels to delta supervision reduces mean gamma RMSE by 60% and 59%, while value error changes by 4.4% at both regimes. The relative curvature improvement is therefore much larger than the change in value fit in this comparison. The gamma improvement is 2.0 reference envelopes at the interpolated regime and 0.5 at the extrapolated regime: it is resolved only at the former. The larger baseline-to-rung-2 improvement is resolved at both. This conditional comparison neither establishes a general preference for gamma labels nor changes the registered parity failure.

3.3 The gain is not confined to the supervised derivative

Rung 2 adds only gamma labels to rung 1, yet RMSE in the unsupervised variance sensitivity $\partial_v u$ falls from $2.66/1.73$ to $1.59/1.28$, a 40% and 26% reduction. Delta error also falls two- to three-fold. The variance-sensitivity gaps are 9.9 and 3.5 reference envelopes, so cross-target transfer is resolved at both regimes. One possible explanation is coupling through the PDE residual, which contains both $\partial_{SS} u$ and $\partial_v u$. That mechanism remains untested; varying the residual and label terms independently would help evaluate it.

What the nested design establishes. This comparison measures the effect of adding gamma labels when delta labels and the shared residual are already present. It does not separate their interactions or establish that gamma is the best next target. Missing configurations include delta plus variance labels and gamma labels alone. These are targeted follow-ups; a full factorial design would require all combinations of the three label types. The transfer result therefore applies to this sequence of added labels.

3.4 The third target yields a resolved gain primarily on itself

Adding variance-sensitivity labels in rung 3 reduces that target's RMSE from $1.59/1.28$ to $0.73/0.52$, differences of 8.0 and 6.0 reference envelopes. Delta also improves at the interpolated regime by 2.2 envelopes, while the gamma improvement is unresolved. At extrapolation, mean delta RMSE increases from 0.003399 to 0.003784, and gamma RMSE in units of 10^{-3} from 0.641 to 0.823, but the differences are only 0.97 and 0.09 envelopes. We therefore cannot conclude that either derivative degrades. All three supervised arms have gamma errors below its reference uncertainty

there (Appendix B). Only the newly supervised target improves beyond the reference uncertainty at both regimes.

The registered label-noise experiment also fails to demonstrate a monotone dose-response. In the confirmatory hedging comparison, adding gamma labels to delta supervision increases loss by a statistically nonzero amount below the 2% negligibility floor. These are distinct downstream tests, reported with their original verdicts in Table 3; they do not identify the mechanism behind the derivative-accuracy results.

3.5 Controls for label information

The pre-registered controls use a curvature penalty without labels, shuffled labels, or additional value data. Against the baseline's held-out gamma relative RMSE of 0.410, the penalty gives 0.982 and shuffled labels 0.408; neither reproduces the supervised improvement. The penalty weight was not tuned. The price-only sweep uses $4{,}096$–$20{,}480$ rows, fewer than the main arms' $23{,}623$, and a residual weight 100 times larger. Its validation gamma relative RMSE changes from 0.647 to 0.596; doubling width gives 0.709 ± 0.061. Because these training conditions differ, the sweep cannot establish whether extra value labels can substitute for derivative labels (Appendix D).

3.6 Hedging performance and the registered endpoint

The confirmatory hedging test fails: the relative improvement is 0.02% against a 10% requirement. The paired 95% interval for the *loss difference*, $[-0.0242, +0.0242]$, includes zero. Together with the model-class departures in Section 3.1, this limits the downstream claim: large improvements in derivative accuracy do not establish a consistent hedging advantage specific to derivative supervision in the evaluated settings.

A separate, exploratory zero-cost comparison considers the same coefficient shift. The fully supervised model's hedging loss is 0.7% above that of the same rule using exact Heston delta. Of its remaining gap to an analytic minimum-variance hedge (Föllmer and Schweizer, 1991; Hull and White, 2017), 96.7% is associated with the choice of hedging rule. Models trained with gamma labels reconstruct that hedge from their own derivatives to within the previously specified 2% tolerance; the closest without gamma labels falls 3.3% short (Appendix D). These comparisons are exploratory and leave the registered failure unchanged. At 1% cost, even the residual-only baseline beats exact Heston delta, which is therefore not a lower bound on loss for this task.

Accuracy rankings depend on the evaluation region. Restricting delta RMSE to the hedged contract's region reverses the two baselines: at the interpolated anchor, the residual-only model is better on the full grid (0.0537 versus 0.0881), but worse regionally (0.0937 versus 0.0409). The reversal holds in all five seeds at both anchors (Table 12). This exploratory metric is not trajectory-weighted and does not establish a general predictor of hedging performance.

4 What transfers, and what does not

Derivative supervision improves accuracy at held-out Heston coefficients, including a derivative that was not directly supervised. Better approximation of the assumed model, however, does not guarantee better decisions when that model is wrong. Under the tested departures and costs, we find no consistent hedging advantage that can be attributed specifically to derivative labels.

The study covers one two-state PDE, a small network, simulated paths and selected departures, with no real-market validation. Derivative-accuracy comparisons use five seeds; ladder hedging comparisons use ten. Labels and reference share a solver family, so solver agreement cannot exclude common bias. Near zero variance, boundary treatment may affect variance sensitivity. No operator-learning baseline or adaptive residual weighting was tested. The fixed hedge need not be optimal at each transaction cost (Whalley and Wilmott, 1997).

Broader validation should test other PDEs, label-noise levels and combinations of derivative labels. A factorial design would help distinguish each label's effect from its interactions with the others. Whether supervising energy curvature also improves force accuracy in molecular models remains untested here. The practical lesson is to measure both derivative error and decision quality, stating where errors are evaluated, how uncertain the reference is, and which costs the decision incurs.

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Declaration of AI-assisted technologies

The authors used generative AI and AI-assisted tools for manuscript drafting, language refinement, and code drafting and debugging. The authors directed the project, reviewed and verified the manuscript, the code, and all reported results, and take full responsibility for the final manuscript.

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- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
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- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 2 gives the design and the validation-only selection rule; Tables 6 and 7 give the architecture, optimizer and schedule, batch and collocation sizes, label artifact, seeds, and the per-arm label flags, together with the residual weight and price-point budget each arm actually carried, including the one control that departs from the shared configuration.

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   - The answer [N/A] means that the paper does not include experiments.
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7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Section 3 states that path-bootstrap intervals are conditional on the fitted models and that exploratory comparisons are unadjusted for multiplicity. Section 3.6 reports the confirmatory interval; Appendix D gives the paired-bootstrap procedure and identifies point-estimate-only comparisons. Appendix B reports reference uncertainty and unresolved orderings.

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   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.

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   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: Table 7 reports per-arm wall clock, parameter count, derivative-evaluation count and peak process memory, and Table 6 names the worker type as single-process CPU. The processor model and a project-wide compute total, including preliminary and discarded runs, were not recorded, so the paper does not meet this item's full accounting standard.

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A Pre-registered acceptance block and amendment record

The decision-bearing portions below are transcribed from `heston_benchmark_v6.yaml`, the contract consumed by the frozen label and training artifacts. Long audit comments are omitted, but key names and values are unchanged. The label artifact is frozen at tag `v6-labels-20260812`.

```
acceptance_thresholds:
  confirmatory_cell_pass: "misspec delta CVaR95 improvement >=10% relative AND paired-bootstrap 95% CI excludes 0
      (combined,1% TC,baseline,rung3 vs standard_pinn)"
  confirmatory_cell_rel_min: 0.10
  headline_scale_free: fraction_of_baseline_to_oracle_gap_closed
  order_attribution: "rung2 beats rung1 at the cell, CI excludes 0; failure = honest null"
  dose_response:
    criteria: "Spearman rho > 0 (shape confirmed via isotonic fit) AND one-sided seed-bootstrap tail probability P(rho<=0)
      < bootstrap_tail_prob_max"
    bootstrap_tail_prob_max: 0.05
    interpretation: "monotonicity not demonstrated is not the claim that the response is flat"
  ood_gamma_rmse_reduction_min: 0.15
  ood_vega_rmse_reduction_min: 0.15
  price_parity_within: 0.10
  in_model_hedging: NOT_PASS_FAIL
  sakuma_null_rel_tol: 0.02
  seeds: {default: 5, confirmatory_cell: 10}
  tail_claim_requires: paired_bootstrap_over_CRN_paths_with_seed_variance_separated

lambda_selection:
  lambda_delta: {value: 1.0, fixed: true, swept: false}
  lambda_pde:
    source_arm: standard_pinn
    candidates: [0.0, 0.01, 0.1, 1.0]
  lambda_gamma:
    source_arm: rung3_delta_gamma_vega
    candidates: [0.3, 1.0, 3.0]
  lambda_vega:
    source_arm: rung3_delta_gamma_vega
    candidates: [0.3, 1.0, 3.0]
  tune_on: validation_only
  robustness_row:
    description: "also report the confirmatory contrast at the rung3-sourced lambda_pde"
    status: robustness_result_not_a_second_confirmatory_test
```

The selected weights were $(\lambda_{\text{pde}}, \lambda_D, \lambda_G, \lambda_v) = (0.01, 1, 1, 1)$. The registered alternative sourced from rung 3 selected $\lambda_{\text{pde}} = 0$ and was reported as a robustness row.

Amendment record. The contract evolved through four documented, pre-run amendments. All preceded training and model evaluation; amendment 3 used a production-scale *protocol diagnostic* to correct the clipping rationale, but no trained-arm outcome.

Table 2: Contract amendments before the frozen label artifact. "Pre-eval" means before any trained-arm evaluation.

ID	Date	Pre-eval	Reason and effect
A1	2026-07-31	yes	Declared formerly implicit thresholds; recorded the realised 43-step clock; distinguished "monotonicity not demonstrated" from flatness; fixed validation-only weight sourcing.
A2	2026-08-04	yes	Made the 10% confirmatory threshold numeric; separated absent from broken evaluations in the verdict vocabulary; declared the corruption-noise ladder of the pre-training detectability diagnostic, and its region of validity.
A3	2026-08-05	yes	Corrected the measured clipping direction in that same diagnostic, demoted its 0.20 noise rung to diagnostic-only, and fixed the extension contingency. The triggering data were clip measurements, not arm outcomes.
A4	2026-08-05	yes	Declared the 20% parameter-point validation split, 64 state–contract–time draws per parameter point, and full Monte Carlo coverage.

Table 3: Registered endpoints and their outcomes. "Not demonstrated" is the contract's own wording for a dose-response that fails its monotonicity criterion, which is not the claim that the response is flat. The in-class check is declared `NOT_PASS_FAIL`: it flags a registered expectation motivated by the similar pricing and Greek accuracy of the supervised models in Sakuma (2026), not a hedging null established by that paper. Intervals below are in loss-difference units; the dose probability is a one-sided seed-bootstrap tail probability.

Endpoint	Declared	Observed	Verdict
Confirmatory: misspecified J, rung 3 vs baseline	$\geq 10\%$ relative, CI excludes 0	$+0.02\%$, CI $[-0.0242, +0.0242]$	fail
Order attribution: rung 2 beats rung 1	CI excludes 0	$+0.0574$, CI $[+0.0527, +0.0621]$; -1.2% relative	fail (reversal)
Dose-response monotonicity	$\rho_S > 0$ and $P(\rho_S \leq 0) < 0.05$	$\rho_S = 0.029$, $P = 0.074$	not demonstrated
Held-out $\partial_{SS}u$ and $\partial_v u$ RMSE at value parity	$\geq 15\%$ each, value parity within 10%	reductions 0.864–0.930; value improved 82–91%	fail (on parity)
In-class consistency check	not pass/fail	$+27.2\%$ gap at zero cost	flag
Mechanism rule	zero-cost gap or cost/-turnover pattern	zero-cost gap; later controls do not isolate a Gamma-specific cause	channel (i), qualified
Departure sweep: locate a favourable regime	—	none at the 1% tier	no decisive regime

B Reference construction and per-derivative solver spread

Section 2 reports agreement across available numerical legs. For derivatives, the measured spread is a diagnostic scale for arm comparisons; it is not a bound on reference error or a guarantee of an ordering.

Construction, and which legs run where. Leg 1 analytically differentiates the trap-free Heston P_1/P_2 Fourier integrands (Heston, 1993; Albrecher et al., 2007) before Gauss–Legendre quadrature. Leg 2 differentiates a Fang–Oosterlee cosine-expansion pricer (Fang and Oosterlee, 2008) with fourth-order central stencils and a three-step Richardson sweep. It separately implements the same ϕ_2 formula as leg 1, so their agreement cannot exclude a shared formula error. Leg 3 is Monte Carlo, pathwise for the first derivative and mixed pathwise/likelihood-ratio for curvature, with antithetic pairs and a control variate; it returns standard errors and has no estimator for the mixed derivative. Leg 4 is a Craig–Sneyd ADI solve (Craig and Sneyd, 1988), routed only where the variance

process is stressed. The interpolated anchor (Feller ratio 1.04) draws all four legs and a four-fold Monte-Carlo path multiplier, $800{,}000$ paths; the extrapolated anchor (Feller ratio 1.78) draws three, $200{,}000$ paths, and has no ADI leg. The legs have distinct numerical procedures but do not provide four fully independent certifications.

Per grid point the reference value is the median across legs and the reported uncertainty is the largest pairwise absolute disagreement, folded with the Monte-Carlo standard error; a point whose largest disagreement exceeds 10^{-3} times a wing-floored scale, plus $3\sigma_{\mathrm{MC}}$ for any pair involving the stochastic leg, is masked. The rule was declared before training.

Deterministic spread and stochastic uncertainty. Table 4 reports RMS disagreement among deterministic legs alongside the broader envelope. At the extrapolated anchor the two characteristic-function legs agree to 1.6×10^{-8} on $\partial_S u$ and 3.3×10^{-8} on $\partial_{SS} u$ in RMS. This agreement cannot rule out their shared bias. The curvature envelope is 2.65 times larger there despite the interpolated anchor's lower Feller ratio. The stored Monte-Carlo standard errors have ratio 2.63; a nominal $n^{-1/2}$ adjustment for the fourfold path-budget difference reduces this to 1.32. This is a scaling diagnostic, not an equal-budget rerun or a bias/convergence certificate. The earlier pointwise MC-dominance percentages are omitted because an audit did not consistently reproduce them under explicit dominance definitions.

Table 4: Per-derivative reference uncertainty at the two evaluation anchors, in the units of Table 1 (curvature $\times 10^{-3}$), as root mean square over the retained points each arm is scored on. "Envelope" is the quantity the paper reports: largest pairwise leg disagreement folded with the Monte-Carlo standard error. "Deterministic" is the largest pairwise disagreement among the non-stochastic legs alone, recomputed on the frozen grids. The ratio is envelope divided by deterministic spread; neither quantity bounds common solver bias.

Regime	Derivative	envelope	deterministic	ratio
interpolated	$\partial u/\partial S$	0.000276	6.96×10^{-6}	40
(4 legs, 8×10^5 paths)	$\partial^2 u/\partial S^2$	0.768	0.00156	491
	$\partial u/\partial v$	0.1083	0.000656	165
extrapolated	$\partial u/\partial S$	0.000397	1.61×10^{-8}	2.5×10^4
(3 legs, 2×10^5 paths)	$\partial^2 u/\partial S^2$	2.035	3.31×10^{-5}	6.1×10^4
	$\partial u/\partial v$	0.1270	5.07×10^{-6}	2.5×10^4

Limits of the resolution heuristic. The reference is a median across available legs. Its agreement with characteristic-function values does not prove truth or independence. Even a valid RMS reference-error bound ϵ would allow a difference of two RMSEs to move by up to 2ϵ; measured solver spread need not be such a bound. Exceeding one envelope therefore supplies a sensitivity diagnostic, not a one-sided guarantee. Falling below it is likewise not evidence of equality.

Which contrasts in Table 1 survive. Table 5 divides each arm-to-arm RMSE difference by the envelope for that derivative and anchor. The convention follows the study's registered `improvement_to_oracle_noise` statistic; it is a resolution heuristic, not a hypothesis test, and it ignores seed dispersion, which is reported separately in Table 1's source.

Table 5: Arm-difference divided by the reference envelope of Table 4. Positive means the second arm is more accurate. A magnitude below 1 falls below the declared resolution heuristic; this is not an inferential significance threshold.

	interpolated			extrapolated		
Contrast	$\partial_S u$	$\partial_{SS} u$	$\partial_v u$	$\partial_S u$	$\partial_{SS} u$	$\partial_v u$
baseline → rung 1	156.2	6.9	67.1	86.8	2.2	45.0
baseline → rung 3	184.7	9.1	85.0	97.3	2.6	54.5
rung 1 → rung 2	26.2	2.0	9.9	11.4	*0.5*	3.5
rung 2 → rung 3	2.2	*0.2*	8.0	*−0.97*	*−0.1*	6.0

Three readings follow, and Sections 3.1–3.4 are written to them.

The headline survives everywhere. Every baseline-to-supervised contrast exceeds the envelope at both anchors, by factors from 2.2 to 185. The 82 to 93% reductions exceed this diagnostic scale, conditional on the reference.

The curvature rung is resolved at one anchor only. The rung-1-to-rung-2 curvature gap is 2.0 envelopes at the interpolated hole and 0.5 at the extrapolated one. Section 3.2 claims the increment at the interpolated anchor and declines to read its extrapolated magnitude finely. The transfer result of Section 3.3 lives in the $\partial_v u$ column and is resolved at both anchors, at 9.9 and 3.5 envelopes.

The two extrapolated reversals are not resolved, and the appendix says so plainly. The flagged risk was that Section 3.4's non-monotonicity rests on $0.641 \to 0.823$ on $\partial^2 u/\partial S^2$ and $0.003399 \to 0.003784$ on $\partial u/\partial S$. Those gaps are 0.09 and 0.97 envelopes. Both sit inside the reference uncertainty, so this benchmark cannot call either a degradation, and Section 3.4 does not. What survives there is the resolved gain on the derivative rung 3 adds, at 8.0 and 6.0 envelopes.

A sharper form of the same limit: at the extrapolated anchor the curvature RMSE of rungs 1, 2 and 3 is 0.77, 0.31 and 0.40 times the envelope. All three errors fall below the envelope, so fine orderings remain conditional on the reference rather than independently certified. Only the gap from the unsupervised arms, which are 3.0 and 3.5 envelopes away, is.

Masking: rate, location, and one correction. The union mask over all five reference quantities removes 17.11% of the interpolated grid (3,704 of 21,648 points) and 9.23% of the extrapolated grid (1,999 of 21,648). Per quantity the interpolated rates are 0.008 (value), 0.038, 0.073, 0.080 and 0.116 for the first derivative, curvature, second state and mixed derivative; the extrapolated rates are 0.016, 0.014, 0.055, 0.044 and 0.052. Table 1 scores every arm on the same union mask.

The mask is a relative-tolerance artefact and sits where that tolerance is hardest to meet. It concentrates overwhelmingly at the shortest maturity, 94.3% of the $\tau = 0.04$ slice at the interpolated anchor and 78.7% at the extrapolated one, against 0.077 to 0.102 beyond $\tau = 0.45$ at the interpolated anchor and 0.000 to 0.020 beyond $\tau = 0.29$ at the extrapolated one; and it peaks just out of the money rather than deep out of it, 47.5% in the $S/K \in [0.75, 0.9)$ band at the interpolated anchor against 18.1% below 0.75 and 6.8% above 1.3. In curvature the pattern is U-shaped, heaviest in the lowest decile, where a relative tolerance on $|\partial^2 u/\partial S^2| \leq 3 \times 10^{-12}$ demands absolute agreement no solver delivers, and in the highest.

The extrapolated anchor is less masked, 9.23% versus 17.11%. The fourth leg at the interpolated anchor introduces additional pairwise checks. These masks are selective; shared support makes arm comparisons consistent on retained points but does not establish performance over excluded states.

C Training configuration and excision protocol

The main ladder and baseline arms share architecture, optimiser, data and step budget; their loss terms differ as listed below. The extra-value-data sweep is an exception. Table 6 states the shared budget once and Table 7 the per-arm difference, so that the identical-configuration claim of Section 2 is checkable rather than asserted.

Excision. Training coefficients are a seeded Latin hypercube on $\kappa \in [1, 4]$, $\theta \in [0.02, 0.12]$, $\xi \in [0.20, 0.60]$, $\rho \in [-0.80, -0.20]$, $v_0 \in [0.01, 0.12]$, rejecting any draw with $2\kappa\theta/\xi^2 < 0.40$ (an oracle-trust cutoff, not the Feller condition). Around each of the five named regimes a ball is excised: in coordinates range-normalised to the unit cube, every draw within Euclidean distance $0.10\sqrt{5} = 0.2236$ of an anchor, that is 10% of the cube diagonal, is rejected. Realised: no sampled point falls inside any ball, the nearest lying at 1.03 to 1.64 times the radius, minimum normalised anchor distance 0.2310 to 0.3670; minimum realised Feller ratio 0.4061.

The validation split is 20% of *coefficient points*, not of rows, so a held-out regime cannot leak into validation through a shared coefficient draw. Both evaluated regimes are therefore absent from training and from every selection score. The interpolated anchor $(\kappa, \theta, \xi, \rho, v_0) = (1.5, 0.04, 0.34, -0.70, 0.02)$ sits inside the sampled box in every coordinate and is a hole punched in the support. The extrapolated anchor $(2.0, 0.04, 0.30, -0.90, 0.04)$ has $\rho = -0.90$ against a sampled range whose lower bound is -0.80 and whose minimum realised draw is -0.7989: the ball

Table 6: Shared configuration. Every entry is identical across the ladder and baseline arms; the extra-value-data control of Table 7 departs from it in the weight, row-budget and selection settings described below.

Component	Setting
Inputs	$(S, K, \tau, \kappa, \theta, \xi, \rho, v_0)$, affinely mapped to $[-1, 1]$ by fixed contract ranges inside the forward pass with no clamp, so evaluated regimes outside those ranges extrapolate rather than saturate
Network	4×64 `tanh` multilayer perceptron, 13,121 parameters, `float32`; no hard-constrained ansatz, terminal treatment identical across arms
Optimiser	Adam, learning rate 3×10^{-3} cosine-annealed to 3×10^{-5}, gradient-norm clip 1.0
Budget	20,000 steps, minibatch 1,024 label rows, 4,096 residual collocation points redrawn each epoch (834 epochs); fixed-step mode (called matched-epoch in code), early stopping disabled
Labels	23,623 training and 5,936 validation rows from the frozen artifact `v6-labels-20260812`: 512 coefficient draws crossed with 64 shared (S, K, τ) triples, mask rate 0.098
Loss normalisation	supervised terms divided by second moments computed once over the retained training split; the Gamma scale uses true consensus `gamma_ref`, shared across arms. Without it the curvature term sits about six orders of magnitude below the value term at equal relative error and is numerically inert
Weights	$(\lambda_{\mathrm{pde}}, \lambda_1, \lambda_2, \lambda_v) = (10^{-2}, 1, 1, 1)$, selected on validation only, sourced per Appendix A
Seeds	5 per arm; 10 on the ladder arms and the residual-only baseline
Hardware	CPU, single process; per-arm wall clock and peak memory in Table 7. Processor model and a project-wide compute total were not recorded

Table 7: Arms: label flags and measured cost. *res.* is the physics residual; ∂_S, ∂_{SS}, ∂_v are the three label terms. Wall clock is the per-seed mean; peak memory is the seed-42 measurement. Derivative evaluations count first- and second-order AD calls, rather than only double-backward passes. $*$ The extra-value-data control uses residual weight 1 instead of 10^{-2} and 12,288 retained rows instead of 23,623. Its checkpoint and run log have conflicting configurations; its cost figures are withheld.

Arm	res.	∂_S	∂_{SS}	∂_v	seeds	wall (s)	deriv. evals	peak (MB)
value-only feedforward	–	–	–	–	5	17	0	383
residual-only baseline	✓	–	–	–	10	377	60,120	978
rung 1, $+\partial_S u$	✓	✓	–	–	10	414	80,160	941
rung 2, $+\partial_{SS} u$	✓	✓	✓	–	10	471	100,200	1009
rung 3, $+\partial_v u$	✓	✓	✓	✓	10	451	100,200	1043
supervised, residual-free	–	✓	✓	✓	5	64	40,080	695
curvature L^2 penalty	✓	–	–	–	5	465	100,200	1267
shuffled labels	✓	✓	✓	✓	5	433	100,200	1095
extra value data*	✓	–	–	–	5	—	—	—

intersects the cube only where it overlaps, and the anchor itself lies outside the sampled support entirely, so that regime is extrapolation and not interpolation into a hole.

The contract grid is $41 \times 33 \times 16$ over $S \in [50, 150]$, $K \in [60, 140]$, $\tau \in [0.04, 1]$ at $r = 0.02$, $q = 0$. Training labels use 64 shared (S, K, τ) triples, giving sparse local coverage near the hedged contract. Absence of a nearby sampled triple does not by itself establish extrapolation outside the training support in these coordinates.

Departure variants. The Bates generating process adds a compound-Poisson component (Bates, 1996) to the baseline anchor $(\kappa, \theta, \xi, \rho, v_0) = (2.0, 0.04, 0.30, -0.50, 0.04)$, over $\lambda_j \in \{0, 0.1, 0.25, 0.5\}$ and $\sigma_j \in \{0.05, 0.10, 0.15\}$ at $\mu_j = -0.10$. The Merton variant (Merton, 1976) replaces the two-state diffusion with a one-state jump-diffusion at $\sigma = 0.20$, $\lambda_j = 0.25$, $\mu_j = -0.10$, $\sigma_j = 0.10$, so the surrogate stays conditioned on a second state the world does not have. Every arm is handed the baseline anchor's coefficients at decision time, never the true ones. The $\lambda_j = 0$ cell is the contract's in-class consistency check and reproduces the Heston cell exactly, 1.7371 against 1.7371. It is the least-departed cell in Table 8 and carries the largest advantage in

it, $+27.7\%$ at zero cost. A second registered check compares the same corner with an expectation of near-indifference motivated by the similar pricing and Greek accuracy of the three supervised models in Sakuma (2026). It *flags*: the gap exceeds that registered expectation. Their accuracy comparison does not itself predict a hedging null in our setting. We retain the flag as a declared consistency check, not a pass/fail endpoint.

The registered severity sweep, read out. Table 8 gives the sweep Section 3.1 summarises. At zero action cost the advantage of the fully supervised arm over the residual-only baseline falls monotonically in *each* departure coordinate, all six one-dimensional chains decreasing, from $+27.7\%$ at the in-class cell to -4.2% at the most departed one, changing sign between $\lambda_j = 0.5,\ \sigma_j = 0.05$ and $\lambda_j = 0.5,\ \sigma_j = 0.10$. Every pooled interval excludes zero except the sign-crossing cell itself. At the registered 1% cost the whole table is harm and no cell is favourable, which is what produced the registered `no_decisive_regime` verdict: that verdict reads only the 1% tier. The ordering is ordinal. We know which cells are more departed, and the advantage decays in that order; we did not compute a distance from the assumed model class, so the axis has no calibrated scale and the table is not a severity-versus-advantage *curve* in the metric sense.

Table 8: Registered departure sweep, delta-only J (lower is better), five seeds, shared trajectories. "Advantage" is the relative improvement of rung 3 over the residual-only baseline; positive favours supervision. † marks the one cell whose pooled paired-bootstrap interval on the difference includes zero. The $\lambda_j = 0$ row is the in-class consistency check, not a departure.

		zero cost		1% cost	
λ_j	σ_j	rung 3	advantage	rung 3	advantage
0 (in class)	0.05	1.737	$+27.7\%$	4.818	-6.9%
0.10	0.05	2.096	$+20.5\%$	5.082	-7.8%
0.10	0.10	2.358	$+17.9\%$	5.330	-7.5%
0.10	0.15	2.699	$+15.6\%$	5.669	-7.0%
0.25	0.05	2.723	$+10.6\%$	5.533	-9.1%
0.25	0.10	3.455	$+7.0\%$	6.207	-8.1%
0.25	0.15	4.384	$+4.9\%$	7.113	-6.5%
0.50	0.05	3.582	$+0.8\%^{\dagger}$	6.205	-11.1%
0.50	0.10	5.057	-3.0%	7.532	-9.6%
0.50	0.15	6.870	-4.2%	9.280	-7.8%
state removed		2.925	$+9.8\%$	5.845	-3.7%

D Supporting exhibits

Value accuracy does not imply derivative accuracy. Table 9 is the exhibit behind Section 1's opening clause. Both unsupervised arms match the reference solution to a few percent while their curvature is wrong by tens of percent, and the ordering on the value is not the ordering on the curvature: the residual-only arm has the better value fit and the worse curvature.

Table 9: Relative RMSE against the reference, at the two held-out coefficient regimes, five seeds. Controlling the value in L^2 leaves curvature error one to two orders of magnitude larger in relative terms.

	interpolated		extrapolated	
Arm	u	$\partial^2 u/\partial S^2$	u	$\partial^2 u/\partial S^2$
residual-only baseline	0.0155	0.4096	0.0128	0.3505
value-only feedforward	0.0328	0.2664	0.0318	0.2206
rung 3 (fully supervised)	0.0014	0.0446	0.0022	0.0498

The information-free controls. Table 10 gives the three controls of Section 3.5 with seeds and dispersion, together with the first-derivative column, which is what separates "no signal" from "objective conflict": the curvature penalty is worse than the baseline on the derivative it never touches

as well as on the one it shapes. The two configuration-matched controls are the ones the argument rests on; the third is discussed below.

Scope of the extra-value-data sweep. The saturation driver uses $\lambda_{\mathrm{pde}} = 1$ rather than the selected 10^{-2}. Its nominal base $N = 4{,}096$ is also smaller than the main arms' actual 23,623 retained rows: even $5N = 20{,}480$ remains below the main budget. The selected $3N$ checkpoint uses 12,288 rows. Thus the level comparison in Table 10 is not matched in weight or data.

Within the sweep, validation curvature relative RMSE changes from 0.647 ± 0.095 to 0.596 ± 0.015 across $1N$–$5N$. The declared 2% rule selects $3N$ (0.605 ± 0.017); its doubled-width control gives 0.709 ± 0.061. This describes the tested protocol and does not rule out optimization limitations. The sweep uses its own validation-selection protocol; its scores are distinct from held-out RMSE. A substitution claim would require matched actual row budgets, residual weights and selection rules. The penalty and shuffled-label controls are configuration-matched, but their failures do not prove a general information bound.

Table 10: Control arms, held-out relative RMSE, mean ± seed standard deviation over five seeds. λ_{pde} and N are the residual weight and actual retained price-row count. Curvature relative RMSE uses the declared central moneyness support; absolute RMSE in Table 1 uses the full retained grid. The extra-data row is not a matched comparison.

			$\partial^2 u/\partial S^2$ rel. RMSE		$\partial u/\partial S$
Arm	λ_{pde}	N	interpolated	extrapolated	interpolated
residual-only baseline	0.01	23,623	0.4096 ± 0.0106	0.3505 ± 0.0158	0.0749
curvature L^2 penalty, no labels	0.01	23,623	0.9822 ± 0.0068	0.9776 ± 0.0079	0.5389
shuffled labels	0.01	23,623	0.4075 ± 0.0343	0.3666 ± 0.0300	0.1360
extra value data (plateau)	1.00	12,288	0.6986 ± 0.0118	0.6147 ± 0.0116	0.2108
rung 3 supervision	0.01	23,623	$\mathbf{0.0446 \pm 0.0028}$	$\mathbf{0.0498 \pm 0.0018}$	**0.0038**

Reconstructing the variance-minimising controller. The combination $\partial_S u + (\rho\xi/S)\,\partial_v u$ is the local-risk-minimising ratio for this state (Föllmer and Schweizer, 1991). Any arm carrying both first derivatives can assemble it from its own outputs with no further reference, so Table 11 is the point at which the field result and the decision result meet. It is exploratory and post-registered, moves no registered verdict, and reuses the frozen checkpoints and the same shared trajectories every other cell faces; ρ and ξ in the correction are the assumed-model values, so no true-process parameter enters. Point estimates only: these are five-seed means against a pre-registered 2% floor, and we claim no interval for them.

Both arms carrying curvature labels land inside the floor, as does the residual-free supervised arm; the two arms without curvature labels miss it by 3.3 and 3.4 percent, and the residual-damaged baseline by 42.8. The reference the registered metric scores against, the exact first derivative alone, is itself 24.7% worse than this controller, which is the sense in which the registered endpoint could not have rewarded accuracy at the accuracy supervision reaches.

Table 11: Delta-only J (lower is better) when the controller is built from each arm's own two first derivatives, on the confirmatory cell's combined perturbation at zero action cost, five seeds, shared trajectories. The gap is relative to the analytic construction; negative beats it. The pre-registered negligibility floor is 2%.

Fields driving the controller	J	gap	within 2% floor
analytic construction	1.6696	—	reference
rung 2 ($+\partial_{SS}u$)	**1.6563**	-0.79%	yes
rung 3 ($+\partial_v u$)	1.6700	$+0.03\%$	yes
supervised, residual-free	1.6793	$+0.58\%$	yes
rung 1 ($+\partial_S u$)	1.7240	$+3.26\%$	no
value-only feedforward	1.7261	$+3.38\%$	no
residual-only baseline	2.3843	$+42.81\%$	no
exact first derivative alone	2.0816	$+24.68\%$	no

Delta accuracy on the full grid and the hedging region. The registered accuracy metric uses the full grid, $S \in [50, 150]$, $K \in [60, 140]$, $\tau \in [0.04, 1]$. The hedge trades one contract, with $K = 100$ and remaining expiry $\tau \in [0.08, 0.25]$. Table 12 restricts the same metric to the corresponding strike, expiry interval and the contract's predefined central range $S/K \in [0.75, 1.30]$. It is a restricted-grid comparison; it does not weight states by how frequently trading trajectories visit them.

As summarised in Section 3.6, delta accuracy reverses the two models without derivative labels: the residual-only model is better on the full grid and worse in the restricted region, in all five seeds at both regimes. Gamma also reverses in the mean, but its full-grid ordering holds in only three of five and four of five seeds. We therefore highlight the delta result in the main text. Across training labels in the same expiry and S/K ranges, pooling strikes, the mask rate is 1.30 times the rate outside. This indicates thinner reference-label coverage in that part of the domain.

Table 12: Delta RMSE for the two models without derivative labels, on the full parameter grid and the restricted hedging region, five seeds. Lower is better. The rankings reverse; the regional metric is not trajectory-weighted.

	interpolated		extrapolated	
Arm	full grid	region	full grid	region
residual-only baseline	**0.0537**	0.0937	**0.0424**	0.0689
value-only feedforward	0.0881	**0.0409**	0.0815	**0.0227**